

QUALITATIVE AND QUANTITATIVE ANALYSIS FOR GENRE INDISTINGUISHABILITY

Tables I:(a) and II:(a) present pair-wise genre sample count for *fiction* and *non-fiction* genre in $\mathcal{D}_{train}$. Each cell in the table represents the number of samples belonging to both the row and column genre class IDs. Tables I:(b) and II:(b) illustrates the co-occurrence ratio between row class ID i in association with column class ID j for *fiction* and *non-fiction* genres, respectively. Each cell contains the ratio of samples belonging to both class IDs i and j to the total number of training samples associated with class ID j . Tables I:(c) and II:(c) reports the misclassification rate by IMAGINE for *fiction* and *non-fiction* genres, respectively. Here, each cell corresponding to row class ID i and column class ID j represents the ratio between the number of testing samples associated with class ID j but wrongly identified as class ID i and the total number of testing samples associated with class ID j that have been misclassified.

Our analysis of Tables I and II suggest that the high co-occurrence of the two genres, coupled with a significant imbalance in the number of training samples associated with one genre without the other, potentially causes the high

misclassification rate. For instance, in Table I, 264 out of 396 *press & media* (ID: 21) books are associated with *sci-fi & fantasy* (ID: 29) genre (refer to Table I:(a)). This leaves only 132 samples that are not categorized as *sci-fi & fantasy*. Conversely, there are 3716 *sci-fi & fantasy* books that do not belong to the *press & media* genre, creating an imbalance between non-*sci-fi & fantasy press & media* and non-*press & media sci-fi & fantasy* samples. This leads to *press & media* genre being misclassified as *sci-fi & fantasy* with a misclassification rate of 0.79. Standard data augmentation methods are limited in their ability to address this imbalance issue, as increasing the number of samples for *press & media* genre, also increases the *sci-fi & fantasy* samples due to inter-genre relation. We observe a similar issues between *humanities* (ID: 14) and *literature* (ID: 16), *animals & wildlife & pets* (ID: 1) and *childrens' book* (ID: 4) (refer to Table I), and *non-fiction* genres *teen & young adult* (ID: 27) and *history* (ID: 13) (refer to Table II). As a result, relatively lower $\mathcal{BA}$ is observed for *press & media*, *humanities*, *animals & wildlife & pets* in *fiction*, and *teen & young adult* in *non-fiction*. Table III showcases the example of misclassified data samples due to genre indistinguishability.

TABLE I: Fiction genre indistinguishability analysis

Class ID	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29		
(a) Sample Count																															
Actual Class ID	1	1002	539	0	817	23	0	153	0	206	143	0	102	41	38	190	362	38	45	158	52	0	0	16	33	27	0	248	0	304	
	2	539	2024	12	1279	32	68	308	44	206	319	24	141	129	54	435	716	46	81	204	137	23	102	154	44	71	53	327	39	882	
	3	0	12	510	0	28	18	0	0	0	0	0	0	154	109	44	160	0	0	213	30	14	0	0	0	106	0	0	14	69	
	4	817	1279	0	2575	104	69	291	44	262	387	0	174	273	72	449	997	70	109	484	242	14	51	74	25	83	189	979	19	876	
	5	23	32	28	104	643	19	9	0	0	0	0	0	62	32	270	32	0	0	137	0	9	19	69	17	0	19	56	0	299	
	6	0	68	18	69	19	473	0	0	0	0	0	0	39	18	54	137	0	0	154	0	0	19	16	18	0	179	16	36	419	
	7	153	308	0	291	9	0	646	0	16	62	38	102	24	40	191	306	16	0	133	16	8	0	92	17	0	0	52	32	103	
	8	0	44	0	44	0	0	0	408	0	0	77	0	103	0	0	191	0	0	268	44	0	52	0	52	0	0	52	0	0	
	9	206	206	0	262	0	0	16	0	713	40	26	0	141	104	40	441	0	0	158	84	0	0	36	48	0	50	76	25	184	
	10	143	319	0	387	0	0	62	0	40	638	80	48	130	0	103	371	31	0	20	48	0	0	97	0	36	0	94	8	28	
	11	0	24	0	0	0	0	38	77	26	80	1128	96	254	139	99	614	89	54	256	151	0	9	376	0	153	0	18	66	159	
	12	102	141	0	174	0	0	102	0	0	48	96	566	30	200	79	365	257	0	84	21	0	0	9	0	0	0	0	142	0	76
	13	41	129	154	273	62	39	24	103	141	130	254	30	2473	138	70	985	108	155	650	382	73	20	574	77	187	65	400	174	506	
	14	38	54	109	72	32	18	40	0	104	0	139	200	138	1169	85	693	226	40	256	287	0	34	97	8	229	26	146	86	339	
	15	190	435	44	449	270	54	191	0	40	103	99	79	70	85	1595	604	25	172	271	38	40	42	100	19	34	40	159	83	454	
	16	362	716	160	997	32	137	306	191	441	371	614	365	985	693	604	4730	378	218	1044	496	67	216	803	219	277	88	899	425	1175	
	17	158	204	213	484	137	154	133	268	158	20	256	84	650	256	271	1044	200	214	3371	258	168	0	379	158	90	236	471	242	1184	
	18	38	46	0	70	0	0	16	0	0	31	89	257	108	226	25	378	752	0	200	29	46	0	117	24	0	8	114	0	144	
	19	45	81	0	109	0	0	0	0	0	0	54	0	155	40	172	218	0	598	214	29	0	64	74	16	0	0	85	0	182	
	20	52	137	30	242	0	0	16	44	84	48	151	21	382	287	38	496	29	258	1517	0	73	352	18	615	0	171	56	427	0	
	21	0	23	14	14	9	0	8	0	0	0	0	0	73	0	40	67	46	0	168	0	396	23	0	16	0	0	14	264	379	
	22	0	102	0	51	19	19	0	52	0	0	9	0	20	34	42	216	0	64	0	73	23	560	0	118	0	63	76	11	379	
	23	16	154	0	74	69	16	92	0	36	97	376	9	574	97	100	803	117	74	379	352	0	0	1778	49	193	12	247	67	423	
	24	33	44	0	25	17	18	17	52	48	0	0	0	77	8	19	219	24	16	158	18	16	118	49	580	0	0	92	0	281	
	25	27	71	106	83	0	0	0	0	0	36	153	0	187	229	34	277	0	0	90	615	0	0	193	0	799	0	51	45	81	
	26	0	53	0	189	19	179	0	0	50	0	0	0	65	26	40	88	8	0	236	0	0	63	12	0	0	657	114	0	766	
	27	248	327	0	979	56	16	52	52	76	94	18	142	400	146	159	899	114	85	471	171	0	76	247	92	51	114	1973	0	94	
	28	0	39	14	19	0	36	32	0	25	8	66	0	174	86	83	425	0	0	242	56	14	11	67	0	45	0	603	94	3980	
	29	304	882	69	876	299	419	103	0	184	28	159	76	506	339	454	1175	144	182	1184	427	264	379	423	281	81	366	766	94	3980	
(b) Cooccurrence																															
Actual Class ID	1	1	0.27	0	0.32	0.04	0	0.24	0	0.29	0.22	0	0.18	0.02	0.03	0.12	0.08	0.05	0.08	0.05	0.03	0	0	0.01	0.06	0.03	0	0.13	0	0.08	
	2	0.54	1	0.02	0.50	0.05	0.14	0.48	0.11	0.29	0.50	0.02	0.25	0.05	0.05	0.27	0.15	0.06	0.14	0.06	0.09	0.06	0.18	0.09	0.08	0.09	0.08	0.17	0.06	0.22	
	3	0	0.01	1	0	0.04	0.04	0	0	0	0	0	0	0.06	0.09	0.03	0.03	0	0	0.06	0.02	0.04	0	0	0.13	0	0	0.02	0.02	0	
	4	0.82	0.63	0	1	0.16	0.15	0.45	0.11	0.37	0.61	0	0.31	0.11	0.06	0.28	0.21	0.09	0.18	0.14	0.16	0.04	0.09	0.04	0.04	0.10	0.29	0.50	0.03	0.22	
	5	0.02	0.02	0.05	0.04	1	0.04	0.01	0	0	0	0	0	0.03	0.03	0.17	0.01	0	0	0.04	0	0.02	0.03	0.04	0.03	0	0.03	0.03	0	0.08	
	6	0	0.03	0.04	0.03	0.03	1	0	0	0	0	0	0	0.02	0.02	0.03	0.03	0	0	0.05	0	0	0.03	0.01	0.03	0	0.27	0.01	0.06	0.11	
	7	0.15	0.15	0	0.11	0.01	0	1	0	0.02	0.10	0.03	0.18	0.01	0.03	0.12	0.06	0.02	0	0.04	0.01	0.02	0	0.05	0.03	0	0	0.03	0.05	0.03	
	8	0	0.02	0	0.02	0	0	0	1	0	0	0.07	0	0.04	0	0	0.04	0	0	0.08	0.03	0	0.09	0	0.09	0	0	0.03	0	0	
	9	0.21	0.10	0	0.10	0	0	0.02	0	1	0.06	0.02	0	0.06	0.09	0.03	0.09	0	0	0.05	0.06	0	0	0.02	0.08	0	0.08	0.04	0.04	0.05	
	10	0.14	0.16	0	0.15	0	0	0.10	0	0.06	1	0.07	0.08	0.05	0	0.06	0.08	0.04	0	0.01	0.03	0	0	0.05	0	0.05	0	0.05	0.01	0.01	
	11	0	0.01	0	0	0	0	0.06	0.19	0.04	0.13	1	0.17	0.10	0.12	0.06	0.13	0.12	0.09	0.08	0.10	0	0.02	0.21	0	0.19	0	0.01	0.11	0.04	
	12	0.10	0.07	0	0.07	0	0	0.16	0	0	0.08	0.09	1	0.01	0.17	0.05	0.08	0.34	0	0.02	0.01	0	0	0.01	0	0	0	0	0.07	0	0.02
	13	0.04	0.06	0.30	0.11	0.10	0.08	0.04	0.25	0.20	0.20	0.23	0.05	1	0.12	0.04	0.21	0.14	0.26	0.19	0.25	0.18	0.04	0.32	0.13	0.23	0.10	0.20	0.29	0.13	
	14	0.04	0.03	0.21	0.03	0.05	0.04	0.06	0	0.15	0	0.12	0.35	0.06	1	0.05	0.15	0.30	0.07	0.08	0.19	0	0.06	0.05	0.01	0.29	0.04	0.07	0.14	0.09	
	15	0.19	0.21	0.09	0.17	0.42	0.11	0.30	0	0.06	0.16	0.09	0.14	0.03	0.07	1	0.13	0.03	0.29	0.08	0.03	0.10	0.08	0.06	0.03	0.04	0.06	0.08	0.14	0.11	
	16	0.36	0.35	0.31	0.39	0.05	0.29	0.47	0.47	0.62	0.58	0.54	0.64	0.40	0.59	0.38	1	0.50	0.36	0.31	0.33	0.17	0.39	0.45	0.38	0.35	0.13	0.46	0.70	0.30	
	17	0.16	0.10	0.42	0.19	0.21	0.33	0.21	0.66	0.22	0.03	0.23	0.15	0.26	0.22	0.17	0.22	0.27	0.36	1	0.17	0.42	0	0.21	0.27	0.11	0.36	0.24	0.40	0.30	
	18	0.04	0.02	0	0.03	0	0	0.02	0	0	0.05	0.08	0.45	0.04	0.19	0.02	0.88	1	0	0.06	0.02	0.12									

TABLE II: Non-fiction genre indistinguishability analysis

Class ID	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	30	
(a) Sample Count																														
Actual Class ID	1	748	169	0	222	41	0	12	0	306	14	0	12	89	52	97	44	14	10	0	16	0	95	0	369	14	0	65	37	157
	2	169	1444	9	331	203	42	0	143	53	34	103	12	512	237	165	109	26	57	58	58	270	408	125	236	14	8	162	28	259
	3	0	9	614	0	46	197	0	0	0	24	0	0	197	303	38	9	0	0	0	34	44	84	0	58	227	0	8	0	107
	4	222	331	0	878	41	0	8	28	84	170	0	31	214	140	82	52	37	46	0	9	0	152	0	340	61	9	315	0	109
	5	41	203	46	41	806	0	0	17	19	10	47	17	308	274	298	50	0	25	24	16	8	216	11	176	19	0	19	16	187
	6	0	42	197	0	0	483	0	0	11	0	0	0	165	183	11	0	8	12	0	0	69	0	201	28	9	0	0	73	
	7	12	0	0	8	0	0	494	0	72	0	9	183	38	26	26	38	41	9	0	11	0	185	12	29	89	0	8	104	111
	8	0	143	0	28	17	0	0	439	45	0	110	8	52	44	8	19	0	34	0	32	0	290	0	0	47	8	0	0	12
	9	306	53	0	84	19	11	72	45	846	0	0	66	197	132	37	39	11	0	76	17	9	174	0	361	9	55	41	122	196
	10	14	34	24	170	10	0	0	0	0	889	119	142	16	521	33	25	149	84	0	193	0	222	225	54	506	9	80	0	57
	11	0	103	0	0	47	0	9	110	0	119	896	0	210	398	85	216	12	68	47	71	33	223	282	39	133	12	0	9	204
	12	12	12	0	31	17	0	183	8	66	142	0	949	17	381	0	11	456	0	0	92	0	400	11	221	506	16	0	0	84
	13	89	512	197	214	308	165	38	52	197	16	210	17	3297	1016	250	556	128	258	383	249	412	478	198	590	9	84	266	369	1198
	14	52	237	303	140	274	183	26	44	132	521	398	381	1016	3417	295	594	353	254	103	532	241	632	369	597	783	24	255	92	581
	15	97	165	38	82	298	11	26	8	37	33	85	0	250	295	1112	97	0	37	39	12	104	149	11	77	33	52	40	147	405
	16	44	109	9	52	50	0	38	19	39	25	216	11	556	594	97	1448	34	236	93	126	102	207	265	66	17	0	96	204	575
	17	0	58	0	0	24	0	0	0	76	0	47	0	383	103	39	93	44	41	744	55	126	103	87	58	9	81	0	221	383
	18	14	26	0	37	0	8	41	0	11	149	12	456	128	353	0	34	886	0	44	143	9	241	20	318	361	0	0	0	179
	19	10	57	0	46	25	12	9	34	0	84	68	0	258	254	37	236	0	762	41	52	0	315	0	32	27	0	25	37	289
	20	16	58	34	9	16	0	11	32	17	193	71	92	249	532	12	126	143	52	55	1122	0	135	205	93	416	12	0	36	147
	21	0	270	44	0	8	0	0	0	9	0	33	0	412	241	104	102	9	0	126	0	735	104	42	42	0	11	8	67	278
	22	95	408	84	152	216	69	185	290	174	222	223	400	478	632	149	207	241	315	103	135	104	2495	171	327	462	129	149	72	107
	23	0	125	0	0	11	0	12	0	0	225	282	11	198	369	11	265	20	0	87	205	42	171	877	70	273	12	0	43	162
	24	369	236	58	340	176	201	29	0	361	54	39	221	590	597	77	66	318	32	58	93	42	327	70	1809	148	0	188	41	207
	25	14	14	227	61	19	28	89	47	9	506	133	506	9	783	33	17	361	27	9	416	0	462	273	148	1519	25	16	9	63
	26	0	8	0	9	0	9	0	8	55	9	12	16	84	24	52	0	0	0	81	12	11	129	12	0	25	424	0	69	185
	27	65	162	8	315	19	0	8	0	41	80	0	0	266	255	40	96	0	25	0	0	8	149	0	188	16	0	659	0	151
	28	37	28	0	0	16	0	104	0	122	0	9	0	369	92	147	204	0	37	221	36	67	72	43	41	9	69	0	888	543
	30	157	259	107	109	187	73	111	12	196	57	204	84	1198	581	405	575	179	289	383	147	278	107	162	207	63	185	151	543	2802
	(b) Cooccurrence																													
Actual Class ID	1	1	0.12	0	0.25	0.05	0	0.02	0	0.36	0.02	0	0.01	0.03	0.02	0.09	0.03	0.02	0.01	0	0.01	0	0.04	0	0.20	0.01	0	0.10	0.04	0.06
	2	0.23	1	0.01	0.38	0.25	0.09	0	0.33	0.06	0.04	0.11	0.01	0.16	0.07	0.15	0.08	0.03	0.07	0.08	0.05	0.37	0.16	0.14	0.13	0.01	0.02	0.25	0.03	0.09
	3	0	0.01	1	0	0.06	0.41	0	0	0	0.03	0	0	0.06	0.09	0.03	0.01	0	0	0.03	0.06	0.03	0	0.03	0.15	0	0.01	0	0.04	
	4	0.30	0.23	0	1	0.05	0	0.02	0.06	0.10	0.19	0	0.03	0.06	0.04	0.07	0.04	0.04	0.06	0	0.01	0	0.06	0	0.19	0.04	0.02	0.48	0	0.04
	5	0.05	0.14	0.07	0.05	1	0	0	0.04	0.02	0.01	0.05	0.02	0.09	0.08	0.27	0.03	0	0.03	0.03	0.01	0.01	0.09	0.01	0.10	0.01	0	0.03	0.02	0.07
	6	0	0.03	0.32	0	0	1	0	0	0.01	0	0	0	0.05	0.05	0.01	0	0.01	0.02	0	0	0.03	0	0.11	0.02	0.02	0	0	0	0.03
	7	0.02	0	0	0.01	0	0	1	0	0.09	0	0.01	0.19	0.01	0.01	0.02	0.03	0.05	0.01	0	0.01	0	0.07	0.01	0.02	0.06	0	0.01	0.12	0.04
	8	0	0.10	0	0.03	0.02	0	0	1	0.05	0	0.12	0.01	0.02	0.01	0.01	0.01	0	0.04	0	0.03	0	0.12	0	0	0.03	0.02	0	0	0
	9	0.41	0.04	0	0.10	0.02	0.02	0.15	0.10	1	0	0	0.07	0.06	0.04	0.03	0.03	0.01	0	0.10	0.02	0.01	0.07	0	0.20	0.01	0.13	0.06	0.14	0.07
	10	0.02	0.02	0.04	0.19	0.01	0	0	0	0	1	0.13	0.15	0	0.15	0.03	0.02	0.17	0.11	0	0.17	0	0.09	0.26	0.03	0.33	0.02	0.12	0	0.02
	11	0	0.07	0	0	0.06	0	0.02	0.25	0	0.13	1	0	0.06	0.12	0.08	0.15	0.01	0.09	0.06	0.06	0.04	0.09	0.32	0.02	0.09	0.03	0	0.01	0.07
	12	0.02	0.01	0	0.04	0.02	0	0.37	0.02	0.08	0.16	0	1	0.01	0.11	0	0.01	0.51	0	0	0.08	0	0.16	0.01	0.12	0.33	0.04	0	0	0.03
	13	0.12	0.35	0.32	0.24	0.38	0.34	0.08	0.12	0.23	0.02	1	0.30	0.22	0.38	0.14	0.34	0.51	0.22	0.56	0.19	0.23	0.33	0.01	0.20	0.40	0.42	0.43	0.21	
	14	0.07	0.16	0.49	0.16	0.34	0.38	0.05	0.10	0.16	0.59	0.44	0.40	0.31	1	0.27	0.41	0.40	0.33	0.14	0.47	0.33	0.25	0.42	0.33	0.52	0.06	0.39	0.10	0.21
	15	0.13	0.11	0.06	0.09	0.37	0.02	0.05	0.02	0.04	0.04	0.09	0	0.08	0.09	1	0.07	0	0.05	0.05	0.01	0.14	0.06	0.01	0.04	0.02	0.12	0.06	0.17	0.14
	16	0.06	0.08	0.01	0.06	0.06	0	0.08	0.04	0.05	0.03	0.24	0.01	0.17	0.17	0.09	1	0.04	0.31	0.13	0.11	0.14	0.08	0.30	0.04	0.01	0	0.15	0.23	0.21
	17	0	0.04	0	0	0.03	0	0	0	0.09	0	0.05	0	0.12	0.03	0.04	0.06	0.05	0.05	1	0.05	0.17	0.04	0.10	0.03	0.01	0.19	0	0.25	0.14
	18	0.02	0.02	0	0.04	0	0.02	0.08	0	0.01	0.17	0.01	0.48	0.04	0.10	0	0.02	1	0	0.06	0.13									

TABLE III: Example of misclassified data samples due to genre indistinguishability

(d) 9780671656010: The Adventures of Alyx; Author: Joanna Russ; Publisher: Baen		
	Blurb	Contents: Bluestocking (1967) I Thought She Was Afeard till She Stroked My Beard (1967) The Barbarian (1968) Picnic on Paradise (1968) novel The Second Inquisition (1970)
	Actual Genre	(Fiction, {Humanities, Sci-Fi & Fantasy })
	Predicted Genre	(Fiction, {Humanities, Literature })
(e) 9780445409132: Murder in the Queen's Armes; Author: Aaron Elkins; Publisher: Warner Books		
	Blurb	Physical anthropologist Gideon Oliver is honeymooning in rural England when a side trip to an archaeological site leads to an adventure in scholarly intrigue and murder
	Actual Genre	(Fiction, { Humanities , Literature, Mystery & Thriller & Suspense & Horror & Adventure})
	Predicted Genre	(Fiction, { History , Literature, Mystery & Thriller & Suspense & Horror & Adventure})
(f) 9780789425225: Voices in the Park; Author: Anthony Browne; Publisher: DK Children		
	Blurb	Four people enter a park and through their eyes young readers see different visions, from the bossy woman and the sad man to the lonely boy and the warm young girl, moving from one voice to another and shifting landscapes and seasons.
	Actual Genre	(Fiction, { Animals & Wildlife & Pets , Arts & Photography, Childrens' Book, Family & Parenting & Relationships, Teen & Young Adult})
	Predicted Genre	(Fiction, {Arts & Photography, Childrens' Book, Family & Parenting & Relationships, Teen & Young Adult})
(g) 9780440436836: Homesick: My Own Story; Author: Jean Fritz; Publisher: Yearling		
	Blurb	Jean Fritz was born in China and lived there until 1927, when she was twelve. Young Jean had spent her entire life in China, but her parents' memories of home and letters from relatives in Pennsylvania made her feel that she was American—and homesick for a place she'd never seen! Family photographs and illustrations by Margot Tones show us the real people behind Jean's vivid and unforgettable stories—memories of picnics on the Great Wall, pranks, holidays in the foreign compound, rebellious moments at her British school, close ties to Chinese friends, and how it felt to be called a "foreign devil" and spat upon in the streets of a turbulent China on the eve of revolution. When her family embarks upon its long journey home, Jean is thrilled, but she When she arrives in America at last, will she fit in after growing up on "the wrong side of the world?"
	Actual Genre	(Non-fiction, {Biographies & Memoir, Childrens' Book, History, Teen & Young Adult })
	Predicted Genre	(Non-fiction, {Biographies & Memoir, Childrens' Book, History})
(h) 9780345291363: Elephant Man; Author: Christine Sparks; Publisher: Ballantine Books		
	Blurb	John Merrick had lived for more than twenty years imprisoned in a body that condemned him to a miserable life in the workhouse and to humiliation as a circus sideshow freak. But beneath that tragic exterior, within that enormous and deformed head, thrived the soul of a poet, the heart of a dreamer, the longings of a man. Based on the extraordinary motion picture that captured the heart of America.
	Actual Genre	(Non-fiction, {Biographies & Memoir, History, Literature, Teen & Young Adult })
	Predicted Genre	(Non-fiction, {Biographies & Memoir, History, Literature})

False positives are marked in red, and False negatives are marked in blue